

1. Create a procedure in PL/SQL to find the minimum of three numbers.

```
SQL> SET SERVEROUTPUT ON;
SQL> CREATE OR REPLACE PROCEDURE find_min(
2   a NUMBER,
3   b NUMBER,
4   c NUMBER
5 )
6 IS
7   minimum NUMBER;
8 BEGIN
9   IF a <= b AND a <= c THEN
10      minimum := a;
11   ELSIF b <= a AND b <= c THEN
12      minimum := b;
13   ELSE
14      minimum := c;
15   END IF;
16
17   DBMS_OUTPUT.PUT_LINE('Minimum = ' || minimum);
18 END;
19 /

Procedure created.

SQL> EXEC find_min(10, 25, 15);
Minimum = 10

PL/SQL procedure successfully completed.
```

2. Create a procedure in PL/SQL to check whether a entered number is even number or odd number. in PL/SQL to find the minimum of three numbers.

```
SQL> CREATE OR REPLACE PROCEDURE check_even_odd(
2   n NUMBER
3 )
4 IS
5 BEGIN
6   IF MOD(n, 2) = 0 THEN
7      DBMS_OUTPUT.PUT_LINE(n || ' is Even');
8   ELSE
9      DBMS_OUTPUT.PUT_LINE(n || ' is Odd');
10  END IF;
11 END;
12 /

Procedure created.

SQL> EXEC check_even_odd(10);
10 is Even

PL/SQL procedure successfully completed.
```

3. Write a function in pl/sql to find the factorial of a number.

```
SQL> CREATE OR REPLACE FUNCTION factorial(
2   n NUMBER
3 )
4 RETURN NUMBER
5 IS
6   fact NUMBER := 1;
7 BEGIN
8   FOR i IN 1..n LOOP
9      fact := fact * i;
10  END LOOP;
11
12  RETURN fact;
13 END;
14 /

Function created.

SQL> SELECT factorial(5) FROM dual;

FACTORIAL(5)
-----
120
```

4. Write a function using PL/SQL to check whether a entered year is a leap year or not.

```
SQL> CREATE OR REPLACE FUNCTION check_leap_year(  
2     y NUMBER  
3 )  
4 RETURN VARCHAR2  
5 IS  
6 BEGIN  
7     IF MOD(y, 4) = 0 AND  
8         (MOD(y, 100) <> 0 OR MOD(y, 400) = 0) THEN  
9         RETURN 'Leap Year';  
10    ELSE  
11        RETURN 'Not a Leap Year';  
12    END IF;  
13 END;  
14 /
```

Function created.

```
SQL> SELECT check_leap_year(2024) FROM dual;
```

```
CHECK_LEAP_YEAR(2024)
```

```
-----  
Leap Year
```

5. Write a function in PL/SQL to find the sum of digits of a 5 digit number.

```
SQL> CREATE OR REPLACE FUNCTION sum_digits(  
2     n NUMBER  
3 )  
4 RETURN NUMBER  
5 IS  
6     num NUMBER := n;  
7     total NUMBER := 0;  
8 BEGIN  
9     WHILE num > 0 LOOP  
10        total := total + MOD(num, 10);  
11        num := TRUNC(num / 10);  
12    END LOOP;  
13  
14    RETURN total;  
15 END;  
16 /
```

Function created.

```
SQL> SELECT sum_digits(12345) FROM dual;
```

```
SUM_DIGITS(12345)
```

```
-----  
15
```

6. Write a PL/SQL function that takes an emp_id as input and returns the emp_name of the corresponding employee.

```
SQL> CREATE OR REPLACE FUNCTION get_emp_name(  
2     p_emp_id NUMBER  
3 )  
4 RETURN VARCHAR2  
5 IS  
6     emp_name Employee_EXP8.Name%TYPE;  
7 BEGIN  
8     SELECT Name  
9     INTO emp_name  
10    FROM Employee_EXP8  
11   WHERE EmpID = p_emp_id;  
12  
13    RETURN emp_name;  
14 END;  
15 /
```

Function created.

```
SQL> SELECT get_emp_name(101) FROM dual;
```

```
GET_EMP_NAME(101)
```

```
-----  
Arjun
```

Create a PL/SQL procedure that calculates and displays the total salary of all employees in the Finance department.

```
SQL> CREATE OR REPLACE PROCEDURE total_finance_salary
2 IS
3     total_salary NUMBER;
4 BEGIN
5     SELECT SUM(Salary)
6     INTO total_salary
7     FROM Employee_EXP8
8     WHERE Dept = 'Finance';
9
10    DBMS_OUTPUT.PUT_LINE('Total Finance Salary = ' || total_salary);
11 END;
12 /
```

Procedure created.

```
SQL> EXEC total_finance_salary;
Total Finance Salary = 93000
```

PL/SQL procedure successfully completed.

8 Implement a PL/SQL function that takes an emp_salary as input and returns the number of employees earning more than the given salary.

```
SQL> CREATE OR REPLACE FUNCTION count_above_salary(
2     p_salary NUMBER
3 )
4 RETURN NUMBER
5 IS
6     emp_count NUMBER;
7 BEGIN
8     SELECT COUNT(*)
9     INTO emp_count
10    FROM Employee_EXP8
11    WHERE Salary > p_salary;
12
13    RETURN emp_count;
14 END;
15 /
```

Function created.

```
SQL> SELECT count_above_salary(40000) FROM dual;
```

```
COUNT_ABOVE_SALARY(40000)
-----
4
```

9. Create a PL/SQL function that calculates and returns the average salary of all employees.

```
SQL> CREATE OR REPLACE FUNCTION average_salary
2 RETURN NUMBER
3 IS
4     avg_sal NUMBER;
5 BEGIN
6     SELECT AVG(Salary)
7     INTO avg_sal
8     FROM Employee_EXP8;
9
10    RETURN avg_sal;
11 END;
12 /
```

Function created.

```
SQL> SELECT average_salary FROM dual;
```

```
AVERAGE_SALARY
-----
44333.3333
```

Write a PL/SQL function that takes an emp_dept as input and returns the total number of employees in that department.

```
SQL> CREATE OR REPLACE FUNCTION count_department(  
2     p_dept VARCHAR2  
3 )  
4 RETURN NUMBER  
5 IS  
6     emp_count NUMBER;  
7 BEGIN  
8     SELECT COUNT(*)  
9     INTO emp_count  
10    FROM Employee_EXP8  
11    WHERE Dept = p_dept;  
12  
13    RETURN emp_count;  
14 END;  
15 /
```

Function created.

```
SQL> SELECT count_department('IT') FROM dual;
```

```
COUNT_DEPARTMENT('IT')  
-----  
3
```

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